

Table 1: *National Air Passenger Taxes, 1998–2018*

Country	Tax	Announced	Effective	EUR per departing passenger
Denmark	Passenger tax extended to domestic flights	1997-05-06	1998-01-01	DKK 75 (≈ 10)
Malta	Departure tax doubled	2004-11-24	2005-08-01	Lm 10 increase (≈ 23)
United Kingdom	Air Passenger Duty doubled	2006-12-06	2007-02-01	GBP 5 / 20 increase (≈ 7 / 30)
Netherlands	Ticket tax (repealed July 2009)	2007-02-07	2008-07-01	11.25 / 45
Ireland	Air Travel Tax (repealed April 2014)	2008-10-14	2009-03-30	2 / 10
Germany	Luftverkehrssteuer	2010-06-07	2011-01-01	8 / 25 / 45
Austria	Flugabgabe	2010-10-23	2011-04-01	8 / 20 / 35
Norway	Flypassasjeravgift	2015-12-14	2016-06-01	NOK 80 (≈ 9)
Sweden	Flygskatt	2017-06-08	2018-04-01	SEK 60 / 250 / 400 (≈ 6 / 26 / 41)

Notes: Announcement is the first public government decision (coalition agreement, cabinet or budget statement, parliamentary vote), read from primary sources; effective is the first taxed departure date. Rates are statutory amounts at introduction by distance band (or the increase, for the UK, Denmark and Malta), converted at event-month exchange rates. All nine taxes exempt connecting passengers on a single ticket, so hub transfer traffic is untaxed. Control airports are those of the other coded European countries (Austria, Belgium, Croatia, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom) with no tax event of their own within three years before the announcement and two years after the effective date; Italy is excluded because its surcharge taxes transfer passengers. The Netherlands enters the pooled estimates of Table 2 only through the robustness windows (Appendix Table A1) because its tax lasted twelve months. Sources: national legal gazettes and budget documents.

Table 2: *Effect of a Ticket Tax in the First Year: Seats, Emissions and Network Position*

	(1) All airports	(2) All, seat-weighted	(3) Small	(4) Mid	(5) Large
ln Seats	−0.103** (0.046)	−0.041* (0.021)	−0.240*** (0.066)	−0.016 (0.060)	−0.048 (0.051)
ln CO ₂	−0.104** (0.047)	−0.021 (0.021)	−0.212*** (0.056)	−0.031 (0.066)	−0.064 (0.051)
ln Connectivity (GACI)	−0.016** (0.006)	−0.023*** (0.007)	−0.023*** (0.005)	−0.022*** (0.003)	−0.003 (0.015)
ln Destinations	−0.089** (0.039)	—	−0.135** (0.062)	−0.106*** (0.023)	−0.034 (0.054)
<i>N</i> (monthly)	112,876	112,306		112,876	
<i>N</i> (annual)	9,615	13,569		9,615	
Countries	33	31		33	

Notes: Stacked difference-in-differences over the eight tax increases of Table 1 (Netherlands excluded). Treated: airports of the taxing country; controls: airports of the coded European countries without a tax event in the window; foreign airports within 50–300 km of a treated airport are a separate group (Appendix Table A2) and those within 50 km are dropped. Pre-period: 36 months before the announcement; post-period: first twelve months from the effective date; months between announcement and effective date dropped. Seats and CO₂ (departing flights, LTO plus cruise) are monthly, with airport-by-calendar-month-by-event and month-by-event fixed effects; GACI (Global Airport Connectivity Index, normalised each year, so coefficients are changes in relative network position) and destinations are annual, three pre-years and the first effective year, with airport-by-event and year-by-event fixed effects (seat-weighted GACI: effective-date window of two years before and after, nine events). Size terciles are defined within each treated country by seats in the year before the announcement (median seats 30 thousand, 168 thousand and 2.0 million; the top tercile carries 94% of the treated countries' seats). Column (2) weights by pre-announcement seats and measures the effect on national totals. Standard errors clustered by country in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 3: Event-Time Coefficients: No Pre-Trends before the Tax

	Before		After	
<i>Panel A. Seats, monthly (bins of twelve months; reference: the year before the announcement)</i>				
	[A−36, A−25]	[A−24, A−13]	[E, E+11]	[E+12, E+23]
All treated airports	−0.051 (0.045)	−0.031 (0.021)	−0.124*** (0.032)	−0.153*** (0.040)
<i>Panel B. Connectivity (GACI), annual (reference: the year before the effective year)</i>				
	<i>E − 3</i>	<i>E − 2</i>	<i>E</i>	<i>E + 1</i>
All treated airports	−0.001 (0.008)	0.000 (0.003)	−0.016*** (0.005)	−0.018*** (0.003)
Small	0.004 (0.005)	−0.005** (0.002)	−0.021*** (0.005)	−0.021*** (0.006)
Mid	−0.006 (0.013)	0.007 (0.005)	−0.021*** (0.006)	−0.022** (0.010)
Large	−0.001 (0.012)	−0.001 (0.004)	−0.004 (0.012)	−0.014 (0.013)

Notes: Specification of Table 2 with event-time bins instead of a single post indicator. Panel A: $\ln(\text{seats})$, pooled eight events, bins of twelve calendar months so that seasonality cannot enter, reference bin the twelve months before the announcement (A); post bins count from the effective date (E). Panel B: $\ln(\text{GACI})$, annual, reference the year before the effective year. Border airports enter as a separate group (not shown). Standard errors clustered by country (33). *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 4: How Emissions Fall: Route Exit, Not Fewer or Smaller Flights

	All treated	Small	Mid	Large
<i>Panel A. Exact decomposition of annual CO₂: $\ln \text{CO}_2 = \ln(\text{destinations}) + \ln(\text{flights per destination}) + \ln(\text{CO}_2 \text{ per flight})$</i>				
$\ln \text{CO}_2$ (total)	−0.100** (0.046)	−0.186** (0.079)	−0.065 (0.043)	−0.064 (0.040)
$\ln$ destinations (network margin)	−0.097** (0.039)	−0.161** (0.060)	−0.083*** (0.025)	−0.051 (0.054)
$\ln$ flights per destination (frequency margin)	−0.010 (0.044)	−0.052 (0.110)	0.025 (0.030)	−0.011 (0.039)
$\ln \text{CO}_2$ per flight (aircraft and distance margin)	0.007 (0.022)	0.027 (0.053)	−0.007 (0.017)	−0.002 (0.019)
<i>Panel B. Monthly margins</i>				
$\ln$ flights	—	−0.237*** (0.080)	−0.052 (0.065)	−0.073* (0.042)
$\ln$ seats per flight	—	−0.002 (0.039)	0.036 (0.022)	0.025** (0.011)
Service loss (0/1, at-risk months)	—	0.044** (0.021)	−0.006 (0.007)	−0.003 (0.007)

Notes: Panel A: the three components add up to total CO₂ by construction (annual stacks of Table 2, three pre-years and the effective year; destinations from the GACI panel). Panel B: monthly stacks of Table 2; service loss is a linear probability model on airport-months with scheduled departures twelve months earlier (months with no departure count as zero seats). Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 5: Connectivity and Emissions: More Connected Airports Emit More in Total but Less per Seat-Kilometre

<i>Panel A. Within-airport elasticities to ln GACI, all airports 1996–2019</i>						
	ln CO ₂	ln destinations	ln seats per flight	ln stage length	ln CO ₂ per seat-km	ln CO ₂ per seat
Airport and year FE	4.722*** (0.289)	4.396*** (0.206)	0.488*** (0.087)	0.662*** (0.099)	−0.532*** (0.082)	0.130** (0.058)
Airport and country-by-year FE	3.630*** (0.282)	4.178*** (0.311)	0.512*** (0.072)	0.762*** (0.083)	−0.499*** (0.072)	0.263*** (0.058)
<i>N</i>	79,613					
<i>Panel B. Decomposition of the tax effect: direct effect holding GACI fixed, and the part running through GACI</i>						
	ln CO ₂ (total)			ln CO ₂ per seat-km		
	Total	Direct	Via GACI	Total	Direct	Via GACI
Small airports	−0.186**	−0.116	−0.069	0.027	0.014	0.013
Mid airports	−0.065	0.007	−0.073	0.008	−0.006	0.013
Large airports	−0.064	−0.052*	−0.012	−0.002	−0.004	0.002

Notes: Panel A: coefficients of ln GACI in airport-year regressions with the stated fixed effects; descriptive within-airport associations, not causal effects. Panel B: Gelbach (2016) decomposition on the annual tax stacks of Table 4: the total tax coefficient by size tercile, the coefficient when ln GACI is added as a control (direct), and the difference, which equals the tax effect on ln GACI times the conditional coefficient of ln GACI (3.40 for total CO₂, −0.63 for CO₂ per seat-km). The decomposition is an accounting identity and assumes GACI is the only omitted channel; it is not an instrumental-variables mediation estimate. Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 6: Synthetic Difference-in-Differences by Event: National Totals Barely Move

Event	Donors	National seats		Mean GACI (seat-weighted)	
		SDID	Placebo <i>p</i>	SDID	Placebo <i>p</i>
NLD 2008-07-01	13	−0.076	0.31	0.020	0.54
IRL 2009-03-30	13	−0.159	0.00	0.006	0.92
DEU 2011-01-01	11	−0.005	1.00	−0.009	0.82
AUT 2011-04-01	12	−0.038	0.83	−0.019	0.58
NOR 2016-06-01	14	−0.155	0.07	−0.019	0.38
SWE 2018-04-01	13	−0.009	1.00	−0.047	0.08
GBR 2007-02-01	15	−0.016	0.80	−0.033	0.29
DNK 1998-01-01	16	−0.099	0.25	—	—
MLT 2005-08-01	13	−0.207	0.00	−0.053	0.15

Notes: Country-level synthetic difference-in-differences (Arkhangelsky et al. 2021): unit and time weights on the simplex with the paper’s ridge penalty. Seats: country-by-month ln(seats) de-seasonalised with pre-announcement country-by-calendar-month means, 36 pre-months, first twelve effective months, anticipation months dropped. GACI: seat-weighted country mean of ln GACI, annual, six pre-years, effective year and the next (Denmark has too few pre-years). Donors: coded European countries without a tax event in the window. Placebo *p*: share of donors whose own estimate, treating that donor with the actual treated country removed, is at least as large in absolute value. Malta has two airports.

Table 7: *First-Year Accounting: Emissions Avoided, Revenue Raised and Connectivity Lost, by Airport Size*

	Airports	CO ₂ avoided (kt)	Revenue (m EUR)	EUR per t CO ₂	GACI lost (index units)	Destinations lost
Small airports	72	42	20	484	0.87	43
Mid airports	67	53	138	2,598	1.00	94
Large airports	70	3,836	3,728	972	0.26	168
All treated airports	209	3,932	3,886	988	2.13	304

Notes: Each row applies the size tercile's first-year coefficients (CO₂ from Table 2; GACI and destinations from the annual stacks) to the pooled treated airports' pre-announcement-year CO₂, GACI and destinations. Revenue is the statutory tax at the airport's distance band times remaining passengers, assuming a load factor of 0.80. The large-airport coefficients are not statistically different from zero, so the large-airport and total rows are point estimates with wide uncertainty; the small- and mid-airport rows rest on precisely estimated coefficients. EUR per tonne compares revenue with avoided emissions and is not a welfare measure.

Table A1: *Robustness of the First-Year Seat Effect (Equal Airport Weights)*

Specification	Taxed $\times$ post	<i>N</i>	Countries
Effective-date window ± 12 months, nine events	−0.115*** (0.024)	66,818	27
Effective-date window ± 18 months, nine events	−0.130*** (0.030)	93,915	27
Effective-date window ± 24 months, nine events	−0.128*** (0.036)	116,250	27
Effective-date window ± 36 months, nine events	−0.130** (0.052)	158,773	27
Without the recession-year events (Netherlands 2008, Ireland 2009)	−0.111*** (0.025)	51,894	27
Announcement donut, two pre-years	−0.128*** (0.040)	111,063	27
Donut, airport-specific pre-trends removed	−0.153*** (0.036)	107,904	27
Donut, three pre-years, verified announcement dates (main)	−0.108** (0.050)	95,270	33

Notes: Treated-by-post coefficient of $\ln(\text{seats})$ from separate stacked regressions with the fixed effects of Table 2. Window rows use the effective date with symmetric windows. Donut rows drop the months between announcement and effective date; the pre-trend row removes each airport’s linear trend fitted on the 24 months before the announcement. The first four donut rows were estimated with the announcement dates as coded before verification (Austria 2011-01, Malta 2004-12); the main row uses the verified dates. Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table A2: *Cross-Border Leakage: Seats at Foreign Airports near the Taxed Country*

	(1) Distance rings	(2) Border airport type
Taxed country $\times$ post	−0.108** (0.050)	−0.108** (0.050)
Foreign airports 0–50 km $\times$ post	−0.202* (0.100)	
Foreign airports 50–150 km $\times$ post	−0.019 (0.042)	
Foreign airports 150–300 km $\times$ post	−0.002 (0.030)	−0.002 (0.030)
Foreign hubs (≥ 5 million seats) within 150 km $\times$ post		0.000 (0.025)
Foreign secondary airports within 150 km $\times$ post		−0.037 (0.047)
<i>N</i>	95,270	95,270

Notes: Specification of Table 2, column (1), with the border group split by great-circle distance to the nearest treated airport, or by size within 150 km. The 0–50 km ring holds few airports and shows pre-trends in the event study, so it is excluded elsewhere. The Dutch tax of 2008, analysed separately, is the one case with measurable leakage: seats at foreign airports within 150 km rose 16% while the tax was in force (results available in the replication files). Standard errors clustered by country. *, **, *** denote significance at the 10%, 5%, and 1% levels.